Meester-Loeys Syndrome (MRLS): Comprehensive Disease Characterization

MONDO: MONDO:0010515 · **OMIM:** #300989 · **Gene:** *BGN* (biglycan), Xq28 **Report type:** Literature-based disease knowledge synthesis (no primary patient data files provided) **Evidence base:** Aggregated disease-level resources (OMIM, Orphanet, GeneReviews) and primary literature; the underlying human data derive from small case/cohort series (individual-patient reports aggregated in Meester 2017 [PMID 27632686] and Meester 2024 [PMID 38531898]).

Summary (Answer to the Research Question)

Meester-Loeys syndrome (MRLS) is a rare **X-linked syndromic form of thoracic aortic aneurysm and dissection (TAAD)** caused by **loss-of-function variants in *BGN***, the gene encoding the small leucine-rich extracellular-matrix proteoglycan **biglycan** (P 27632686; 34807424). Hemizygous males present with **early-onset, often catastrophic aortic aneurysm/dissection** together with connective-tissue, skeletal, craniofacial, cutaneous and neurological features that clinically overlap Marfan and Loeys-Dietz syndromes; heterozygous females are more variably and usually more mildly affected. The core mechanism combines **structural weakening of the arterial extracellular matrix (ECM)** (disorganized collagen fibrils, reduced tensile strength) with **increased TGF-β/SMAD2 signaling** in the aortic wall, validated by *Bgn*-null male mice that die from spontaneous aortic rupture (P 17502576; 27632686). Management is extrapolated from heritable-TAAD protocols (imaging surveillance, β-blockers/angiotensin-receptor blockers, prophylactic aortic surgery); no MRLS-specific therapies are approved.

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1. Disease Information

Overview. MRLS is a Mendelian, X-linked syndromic aortopathy/connective-tissue disorder. "*Meester-Loeys syndrome is an X-linked form of syndromic thoracic aortic aneurysm, characterized by the involvement of multiple organ systems... the cardiovascular, skeletal, craniofacial, cutaneous and neurological systems are affected. Clear clinical overlap with Marfan syndrome and Loeys-Dietz syndrome is observed. Aortic dissections occur typically at young ages and are most often observed in males*" (P 34807424).

Key identifiers. - **OMIM:** 300989 (Meester-Loeys syndrome) - **MONDO:** MONDO:0010515 - **Orphanet:** ORPHA:404003 (Meester-Loeys syndrome) - **Gene / OMIM gene:** *BGN* — OMIM 301870; HGNC:1044; NCBI Gene 633; Ensembl ENSG00000182492; UniProt P21810 (PGS1/ biglycan) - **ICD-11:** best mapped under LA87 / heritable connective-tissue and thoracic-aorta disorders (no MRLS-specific code); **ICD-10:** Q87.4 (Marfan-like syndromes) / I71.x (aortic aneurysm/dissection) used pragmatically - **MeSH:** no dedicated descriptor; indexed under "Aortic Aneurysm, Thoracic" + "Genetic Diseases, X-Linked" + "Proteoglycans"

Synonyms / alternative names. Meester-Loeys syndrome; MRLS; **BGN-related thoracic aortic aneurysm and dissection**; X-linked syndromic TAAD (biglycan type); "biglycan-related Meester-Loeys syndrome" (P 38531898).

Data provenance. Disease-level aggregation from OMIM/Orphanet, built on aggregated individual-patient case series (18 probands + 36 relatives to date; P 38531898).

2. Etiology

Primary cause — genetic. MRLS is monogenic, caused by **hemizygous (males) or heterozygous (females) loss-of-function variants in *BGN***. "*We found five individuals with loss-of-function mutations in BGN encoding the small leucine-rich proteoglycan biglycan*" (P 27632686). No infectious or purely environmental cause exists.

Genetic risk factors. - **Causal variants:** *BGN* loss-of-function — nonsense, frameshift, canonical splice-site, and whole-/partial-gene deletions. "*The identified BGN variants were shown to lead to loss-of-function... No (likely) pathogenic missense variants without additional (predicted) splice effects were identified*" (P 38531898) — i.e., pure missense variants are essentially not disease-causing; loss of function is the operative mechanism. - **Modifier genes / contiguous-gene effects:** a deletion extending from *BGN* into the 5′UTR of the neighboring **ATP2B3** gene produced a **more severe skeletal phenotype**, "*possibly explained by expressional activation of the downstream ATPase ATP2B3... driven by the remnant BGN promotor*" (P 38531898). Genes in the shared ECM/TGF-β aortopathy network (*FBN1, TGFBR1/2, SMAD3, TGFB2/3, COL3A1, ACTA2, MYH11*) are candidate genetic-background modifiers (inferred).

Environmental / lifestyle risk factors (disease-triggering, not disease-causing). As in all heritable TAAD, **hypertension and increased hemodynamic/wall stress** promote aneurysm growth and dissection: "*The major risk factors for the disease are increased hemodynamic forces, typically owing to poorly controlled hypertension, and heritable genetic variants*" (P 31066871). **Male sex** is a strong effect modifier (X-linked dosage; see §9). Isometric/heavy resistance exercise, stimulant use, and pregnancy are conventionally avoided/monitored in TAAD (inferred by extrapolation).

Protective factors. In females, **skewed (favorable) X-inactivation** biasing expression toward the wild-type *BGN* allele is the plausible protective modifier explaining milder female phenotypes (inferred). No validated protective germline variant is described. Pharmacological "protection" (β-blockade/ARB) is discussed in §12.

Gene–environment interaction. Genetic ECM/TGF-β vulnerability × mechanical wall stress (blood pressure, exercise, pregnancy) determines dissection timing/severity; the murine model implicates "*gender-related response to stress*" as an interacting determinant of the male-limited catastrophe (P 17502576).

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3. Phenotypes

Phenotype types span **clinical signs/physical manifestations** (cardiovascular, skeletal, craniofacial, cutaneous) and **structural imaging abnormalities**; onset is typically congenital-to-childhood for morphological features and **childhood-to-young-adult for aortic events**. Frequencies below are qualitative/approximate (small cohorts, N≈18 probands).

Cardiovascular (core, most severe). - **Thoracic aortic aneurysm (aortic root/ascending)** — HP:0012727 / aortic root aneurysm HP:0002616. Common, early. Severity severe; progression progressive. - **Aortic dissection** — HP:0002647. "*early-onset aortic aneurysm and dissection*" (P 27632686); "*Aortic dissections occur typically at young ages and are most often observed in males*" (P 34807424). - **Aneurysms/dissections of the wider arterial tree** (not confined to thoracic aorta) — "*aneurysms and dissections in MRLS extend beyond the thoracic aorta, affecting the entire arterial tree*" (P 38531898). HP:0004942 (arterial aneurysm), HP:0025019 (arterial dissection). - **Arterial tortuosity** — HP:0005116 (reported in the Marfan/LDS-overlap spectrum; frequent). - **Congenital heart defects / valvular disease** — including septal defects and valvular anomalies; HP:0001631 (atrial septal defect), HP:0001629 (ventricular septal defect), HP:0001634 (mitral valve prolapse), HP:0001647 (bicuspid aortic valve). Variable frequency.

Craniofacial. - **Hypertelorism** — HP:0000316 ("*Other recurrent findings include hypertelorism*", P 27632686). Frequent. - Additional dysmorphism (e.g., broad forehead/macrocephaly, downslanting palpebral fissures, high palate) — variable; HP:0000256 (macrocephaly), HP:0000494 (downslanting palpebral fissures).

Skeletal / connective tissue. - **Pectus deformity** (excavatum/carinatum) — HP:0000766. Frequent (P 27632686). - **Joint hypermobility** — HP:0001382. Frequent. - **Joint contractures** — HP:0001371 ("*joint hypermobility, contractures*", P 27632686). - **Mild skeletal dysplasia / short stature / scoliosis / arachnodactyly-like features** — HP:0002652 (mild skeletal dysplasia spectrum), HP:0002650 (scoliosis), HP:0001166 (arachnodactyly). Variable; a contiguous *ATP2B3* deletion caused a more severe skeletal phenotype (P 38531898).

Cutaneous. Soft/hyperextensible or translucent skin, striae, easy bruising (connective-tissue fragility) — HP:0000974 (hyperextensible skin), HP:0001075 (atrophic scars/soft skin spectrum). Variable.

Neurological. Reported involvement (e.g., developmental/structural CNS findings in some patients) — the neurological system is listed among affected systems (P 34807424); frequency low/variable.

Quality-of-life impact. No MRLS-specific QoL (EQ-5D/SF-36/PROMIS) data exist. By analogy to Marfan/LDS: lifelong cardiovascular-event anxiety, activity restriction (avoidance of contact/heavy resistance sport), chronic musculoskeletal pain from hypermobility/pectus/scoliosis, and surgical burden reduce QoL; sudden aortic death is the dominant life-limiting concern in males.

4. Genetic / Molecular Information

Causal gene. ***BGN*** (biglycan), **Xq28**; HGNC:1044; NCBI Gene 633; OMIM 301870; UniProt **P21810**. Class I small leucine-rich proteoglycan (SLRP); "*Biglycan is a Class I Small Leucine Rich Proteoglycan (SLRP) that is localized on human chromosome Xq28-ter... Biglycan contains two chondroitin sulfate glycosaminoglycan (GAG) chains attached near its NH2 terminus*" (P 12975603). Paralog/duplication partner: **decorin (DCN)**.

Pathogenic variants. - **Type/class:** predominantly **loss-of-function** — nonsense, frameshift indels, canonical splice-site variants, and **partial/whole-gene deletions** (e.g., chrX:153,502,980–153,530,518del, GRCh38; P 36599284). Larger deletions may extend into contiguous genes (*ATP2B3*). - **Functional consequence: loss of function** / **loss of expression** confirmed by cDNA and Western blot of skin fibroblasts (P 38531898). Pure missense variants without splice effect are **not** established as pathogenic (P 38531898) — argues against a dominant-negative missense mechanism and for haploinsufficiency/absence of protein. - **Classification (ACMG/AMP):** reported variants are pathogenic/likely pathogenic; loss-of-function is a recognized mechanism for *BGN* (PVS1 applicable), supported by segregation. - **Allele frequency:** private/ultra-rare; absent or vanishingly rare in gnomAD (constraint expected for an X-linked LoF disease gene). Genomic coordinates in GRCh38; ClinVar hosts the reported variants. - **Somatic vs germline: germline**. De novo and inherited (X-linked segregation) variants both occur (P 38531898).

Modifier genes / epigenetics. - **ATP2B3** contiguous-gene activation modifies skeletal severity (P 38531898). - **X-chromosome inactivation (XCI)** is the principal epigenetic determinant of female expressivity (skewed XCI → variable phenotype) (inferred). - No disease-specific DNA-methylation/histone signature has been reported.

Chromosomal abnormalities. Structural deletions at Xq28 spanning *BGN* (± *ATP2B3*) are detectable by chromosomal microarray / MLPA / read-depth analysis (P 38531898; 36599284).

Ontology tags. Gene: HGNC:1044 (*BGN*). GO molecular function: GO:0005518 (collagen binding), GO:0050431 (transforming growth factor beta binding), GO:0030020 (extracellular matrix structural constituent conferring tensile strength).

5. Environmental Information

- **Environmental toxins/radiation/infection:** none causal or contributory established; MRLS is monogenic.
- **Lifestyle / hemodynamic factors:** uncontrolled hypertension and high-intensity isometric exercise increase aortic wall stress and dissection risk in heritable TAAD generally ("*increased hemodynamic forces, typically owing to poorly controlled hypertension*", P 31066871). Smoking and stimulant drugs are conventional aggravators of aortic disease (inferred). **Pregnancy** is a high-risk period for aortic dissection in connective-tissue aortopathies (inferred/extrapolated).
- **Infectious agents:** not applicable.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation)

1. A **germline loss-of-function *BGN* variant** (nonsense/frameshift/splice/deletion) at Xq28 **leads to** absent or non-functional **biglycan** protein (loss of function demonstrated by cDNA/Western; P 38531898).

2. Loss of biglycan **results in** defective **collagen fibrillogenesis and ECM assembly** — biglycan normally binds type I collagen and organizes fibrils ("*able to associate specifically with type I collagen fibrils... inhibit collagen fibrillogenesis*", P 8038266) — **leading to** structurally abnormal collagen fibrils and **reduced tensile strength** of the arterial wall (demonstrated in *Bgn*-null mouse aortas by TEM/biomechanics; P 17502576).
3. In parallel (branch point), loss of biglycan's role as a **matrix reservoir/modulator of TGF-β and BMP** ("*less BMP-4 binding, which reduced the sensitivity of osteoblasts to BMP-4*", P 15173106; TGF-β binding, P 9731537) **results in increased/dysregulated TGF-β signaling**, demonstrated in the human aortic wall as **increased nuclear pSMAD2** ("*an increase in nuclear pSMAD2 in the aortic wall*", P 27632686). *(Whether TGF-β activation is upstream driver or downstream compensatory response is not fully resolved — inferred parallel to Marfan/LDS.)*
4. Increased canonical (SMAD2/3) and non-canonical (ERK-mediated, via AT1 receptor) TGF-β signaling **leads to medial degeneration/remodeling** of the aortic media (SMC phenotype change, matrix turnover) — the shared final common pathway of heritable aortopathy (P 25614286; 26607280).
5. Mechanically weakened, remodeled arterial media under hemodynamic stress **results in aortic root/ascending aneurysm**, and, when the intima/media tears, **aortic dissection and rupture** — occurring across the **entire arterial tree** (P 38531898), preferentially and earlier in **males** ("*gender-related response to stress*", P 17502576).
6. Loss of biglycan in **bone/skeletal ECM** independently **leads to** reduced osteoblast differentiation and low bone mass/skeletal features

(P 9731537 ; 15173106), and in **skin/joints** to connective-tissue laxity — producing the non-cardiovascular manifestations.

7. Branch (context-dependent): soluble biglycan is normally a **TLR2/TLR4 danger signal (DAMP)** engaging CD14 (pro-inflammatory) or CD44 (pro-autophagic) (P 30776184 ; 24480070). Its loss may perturb ECM-immune homeostasis; the net contribution to MRLS vascular inflammation is **inferred, not demonstrated**.

Detail by category

- **Molecular pathways:** TGF-β/SMAD2/3 (KEGG hsa04350; Reactome R-HSA-170834), BMP signaling, angiotensin-II AT1-receptor→ERK (non-canonical). CHEBI: TGF-β is a protein (not CHEBI); losartan CHEBI:6541; angiotensin II CHEBI:2719.
- **Protein dysfunction:** loss of function / absent proteoglycan (not aggregation) — UniProt P21810; SLRP leucine-rich-repeat fold (InterPro IPR000372/Pfam LRR).
- **Cellular processes:** vascular smooth-muscle-cell phenotype modulation, ECM remodeling, medial degeneration; osteoblast differentiation defect; possible sterile inflammation/autophagy switch (biglycan DAMP).
- **Tissue-damage mechanism:** mechanical failure of collagen-poor/disorganized media → intimomedial tear, dissection, hemorrhage (P 17502576).
- **Metabolic/biochemical:** no primary metabolic defect; the defect is structural-ECM and growth-factor-signaling.
- **Molecular profiling:** patient iPSCs (P 36599284) and skin-fibroblast cDNA/protein analyses (P 38531898) are available; no large MRLS transcriptomic/proteomic/metabolomic datasets published.
- **GO/CL suggestions:** GO:0030198 (extracellular matrix organization), GO:0032964 (collagen biosynthetic process/fibril organization GO:0030199), GO:0007179 (TGF-β receptor signaling

pathway), GO:0006954 (inflammatory response, branch). Cell types: CL:0000359 (vascular associated smooth muscle cell), CL:0000057 (fibroblast), CL:0000062 (osteoblast).

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7. Anatomical Structures Affected

Organ level (primary). Aorta — especially **aortic root/ascending aorta** (UBERON:0035904 ascending aorta; UBERON:0000947 aorta); **entire arterial tree** including branch/peripheral arteries (UBERON:0001637 artery) (P 38531898).

Heart/valves (UBERON:0000948) with congenital defects. **Secondary/complication level.** End-organ ischemia from dissection (brain, viscera, limbs); hemothorax/hemoperitoneum from rupture (P 17502576).

Body systems. Cardiovascular (primary); musculoskeletal/skeletal; integumentary (skin); craniofacial; nervous system (variable).

Tissue and cell level. Connective tissue / ECM of the arterial wall (media and adventitia), bone, skin, joints. Key cells: **vascular smooth muscle cells** (CL:0000359), **(myo)fibroblasts** (CL:0000057/CL:0000186), **osteoblasts/marrow stromal precursors** (CL:0000062/CL:0000134). SLRPs including biglycan localize to medial and adventitial arterial layers (P 24272803).

Subcellular level. Extracellular matrix / extracellular space (GO:0031012 extracellular matrix; GO:0005615 extracellular space) is the primary compartment; secretory pathway (ER/Golgi glycosylation of GAG chains) is relevant to biosynthesis; **nucleus** implicated via pSMAD2 nuclear translocation (GO:0005634).

Localization / laterality. Aortic disease is **central/axial and typically bilateral-symmetric** in the sense of a midline great vessel; peripheral arterial aneurysms may be multifocal. Skeletal/craniofacial features are typically bilateral.

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8. Temporal Development

- **Onset.** Morphological (craniofacial/skeletal) features are **congenital**; aortic aneurysm can be detected in **childhood**, and **dissection occurs at young ages**, frequently in the first three decades in males ("*Aortic dissections occur typically at young ages*", P 34807424; the mouse model dies within the first 3 months, P 17502576).
- **Onset pattern.** Aneurysm growth is **chronic/insidious**; dissection/rupture is an **acute, catastrophic** event.
- **Progression.** Aortic dilatation is **progressive**; disease course is **chronic and lifelong** with risk of episodic acute events. Progression rate is **variable** and **sex-dependent** (faster/more severe in males; P 38531898).
- **Stages (pragmatic).** (i) pre-aneurysmal/at-risk carrier → (ii) aortic dilatation/aneurysm → (iii) dissection/rupture (end-stage acute event) → (iv) post-surgical residual disease requiring lifelong surveillance.
- **Remission.** No spontaneous remission; **treatment-induced stabilization** (surgery + medical therapy) is the goal.
- **Critical periods / windows of intervention.** Childhood–young adulthood surveillance to intervene **before** dissection; peri-operative timing at diameter thresholds; pregnancy is a critical high-risk window in females.

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9. Inheritance and Population

Epidemiology. Ultra-rare; prevalence unknown, not formally estimated (Orphanet lists no point prevalence). Only **18 probands + 36 variant-harboring relatives** were in the largest cohort (P 38531898). Incidence unquantified.

Inheritance (genetic). - **Pattern: X-linked** (Xq28). Hemizygous males fully/severely affected; heterozygous females variably affected. "*BGN gene defects in humans cause an X-linked syndromic form of severe TAAD*" (P 27632686).

- **Penetrance:** high/near-complete in **males**; reduced in **females**. "*the clinical presentation is more severe and penetrant in males compared to females*" (P 38531898).

- **Expressivity: variable**, especially in females (modulated by X-inactivation). - **Genetic anticipation:** not described (not a repeat-expansion disorder). - **Germline mosaicism:** possible in principle for X-linked LoF disorders (not specifically documented for MRLS). - **De novo variants:** occur; both inherited and de novo variants reported. - **Founder effects / consanguinity:** none reported; variants are largely private. - **Carrier frequency:** not established; expected very low (ultra-rare gene).

Population demographics. - **Sex ratio:** strong **male predominance** of clinically significant disease (16/18 probands male; P 38531898).

- **Ethnic/geographic distribution:** no enrichment; cases reported across multiple countries/ancestries (international cohort, P 38531898).

No variant-specific geographic clustering. - **Age distribution:** young — pediatric to young-adult events predominate.

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10. Diagnostics

Genetic testing (definitive). - **Approach:** molecular confirmation of a **pathogenic *BGN* loss-of-function variant**. Recommended via **multigene heritable-TAAD/aortopathy NGS panels** (including *FBN1*, *TGFBR1/2*, *SMAD3*, *TGFB2/3*, *COL3A1*, *ACTA2*, *MYH11*, *LOX*, *BGN*, etc.), **WES/WGS**, or **single-gene *BGN* sequencing + deletion/duplication analysis** (MLPA/CMA for the structural deletions). "*Extensive analysis at RNA, cDNA, and/or protein level is recommended*" to prove loss of function and resolve splice effects

(P 38531898).

- **CMA / MLPA / read-depth:** needed to detect partial/whole-gene *BGN* deletions (± *ATP2B3*) (P 38531898 ; 36599284). - **RNA/cDNA studies & Western blot of skin fibroblasts:** functionally confirm splice/LoF impact and absence of protein (P 38531898).

- Karyotype/FISH/mtDNA/repeat-expansion testing: not indicated (not applicable).

Clinical / imaging tests. - **Echocardiography (transthoracic)** for aortic root/ascending dimensions — first-line surveillance. - **CT angiography / MR angiography of the whole arterial tree** — essential because disease "*extend[s] beyond the thoracic aorta, affecting the entire arterial tree*" (P 38531898); assesses tortuosity, distal aneurysms. - **Physical exam** for connective-tissue/skeletal/craniofacial signs (hypertelorism, pectus, hypermobility, contractures). - **Histopathology (if tissue available):** medial degeneration with **preserved elastic fibers** but abnormal collagen — a distinguishing feature ("*preservation of elastic fibers and increased TGF-β signaling*", P 27632686); increased nuclear pSMAD2 on IHC.

Biomarkers. No validated circulating MRLS biomarker. Increased aortic-wall pSMAD2 is a tissue marker of TGF-β activation (P 27632686). Circulating TGF-β/TGF-β2 is a candidate (elevated in related ECM aortopathies, P 26607280) — **inferred, unvalidated in MRLS**.

Clinical criteria / differential diagnosis. No standalone diagnostic criteria; diagnosis rests on phenotype + *BGN* variant. **Differential diagnoses:** Marfan syndrome (*FBN1*), Loeys-Dietz syndrome (*TGFBR1/2, SMAD2/3, TGFB2/3*), vascular Ehlers-Danlos (*COL3A1*), and the other X-linked aortopathy **FLNA/filamin-A** (periventricular nodular heterotopia + aortic disease;

P 35819109).

Distinguishing features of MRLS: X-linked male-predominant inheritance, preserved elastic fibers with collagen fibril abnormality, hypertelorism + pectus + hypermobility/contractures.

Screening. **Cascade genetic testing** of at-risk relatives once the familial *BGN* variant is known; **prenatal/preimplantation testing** feasible. No newborn screening exists.

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11. Outcome / Prognosis

- **Survival/mortality.** Prognosis is dominated by **aortic dissection/rupture, a leading cause of early death in affected males**. No formal 5-/10-year survival statistics exist (ultra-rare); the mouse model shows **50% male mortality by 3 months** from aortic rupture (P 17502576), underscoring lethality of untreated severe disease.
- **Disease-specific mortality.** Sudden cardiovascular death from dissection/rupture is the principal disease-specific cause.
- **Morbidity/disability.** Post-dissection sequelae, multiple aortic/arterial surgeries, chronic musculoskeletal disability (scoliosis, pectus, hypermobility pain), and lifelong activity restriction.
- **Recovery.** No cure; timely prophylactic surgery + medical therapy markedly improves outcome (extrapolated from Marfan/LDS).
- **Prognostic factors.** **Male sex, larger/rapidly growing aortic diameter, distal/multifocal arterial involvement, and large deletions (± *ATP2B3*)** predict worse outcome (P 38531898). QoL: no MRLS-specific instrument data.

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12. Treatment

No MRLS-specific approved therapy or trial exists; management is extrapolated from heritable-TAAD/Marfan/LDS guidelines.

Pharmacotherapy (medical aortic protection). - **β-adrenergic blockers** (e.g., atenolol, metoprolol) — reduce dP/dt and wall stress (NCIT: C2019 adrenergic beta-antagonist). Standard of care in heritable TAAD. - **Angiotensin-II type-1 receptor blockers (ARBs), esp. losartan** (NCIT:C61912 losartan; DrugBank DB00678; CHEBI:6541) — reduce TGF-β signaling and aortic growth in Marfan models/patients: "*TGF-β2 levels were reduced after losartan treatment, an angiotensin-II type-1 receptor blocker, known to prevent aortic aneurysm formation*" (P 26607280); clinical-trial context in P 31066871 / 27274304. **Biologically rationalized in MRLS** given increased aortic pSMAD2, but efficacy is **unproven** in MRLS. - **Blood-pressure control** generally to minimize hemodynamic stress (P 31066871). - **Pharmacogenomics:** none specific to MRLS.

Surgical / interventional. Prophylactic aortic root/ascending replacement at diameter thresholds (with valve-sparing where feasible), and repair of dissections/peripheral aneurysms; lifelong surveillance of the **whole arterial tree** given diffuse involvement (P 38531898). NCIT: C157975 (aortic aneurysm repair) / C51826 (aortic surgery).

Advanced / experimental therapeutics. No gene, cell, or RNA therapy in clinical use. Patient-derived **iPSCs** (P 36599284) enable disease modeling and future therapeutic screening. Anti-TGF-β strategies are conceptually relevant but require careful timing (TGF-β neutralization can be beneficial or harmful depending on disease stage; P 25614286).

Supportive/rehabilitative. Physiotherapy for hypermobility/scoliosis, pectus management, pain control, cardiology + medical genetics multidisciplinary care.

Treatment strategy. Genotype-informed, risk-stratified surveillance + medical therapy + timely prophylactic surgery; avoid heavy isometric exercise; high-risk pregnancy management.

13. Prevention

- **Primary prevention:** not possible (monogenic); **genetic counseling** and **reproductive options** (PGT/prenatal testing) prevent transmission (NSGC/ACMG frameworks).
- **Secondary prevention (early detection): cascade genetic testing** of relatives; **serial aortic imaging** (echo + CT/MR angiography) in variant carriers to detect aneurysm before dissection.
- **Tertiary prevention (complication avoidance):** β-blockers/ARBs, blood-pressure control, activity modification, and **prophylactic surgery at threshold diameters** to prevent dissection/rupture (extrapolated from P 31066871).
- **Risk stratification:** males and large-deletion carriers prioritized for intensive surveillance (P 38531898).
- **Immunization / public-health / environmental measures:** not applicable.
- **Counseling:** X-linked recurrence-risk counseling — affected/carrier females transmit to 50% of offspring; affected males transmit the variant to all daughters (obligate carriers) and no sons.

14. Other Species / Natural Disease

- **Taxonomy / model species:** *Mus musculus* (NCBI:txid10090) is the principal species with a described *Bgn*-related aortic phenotype.
- **Orthologous gene:** mouse ***Bgn*** (NCBI Gene 12111; X chromosome), highly conserved with human *BGN*; conserved intron-exon structure shared with decorin (P 12975603).
- **Naturally occurring animal disease:** no well-characterized spontaneous *BGN*-related aortic-dissection disorder in companion animals/wildlife is documented in OMIA (as of this review); the murine phenotype is engineered, though it arose on a specific genetic

background (BALB/cA) revealing strain/background dependence
(P 17502576).

- **Comparative biology / conservation:** the biglycan–collagen–TGF-β ECM axis is evolutionarily conserved; the male-limited murine aortic-rupture phenotype parallels human male predominance, supporting conserved sex-dependent mechanisms (P 17502576).
- **Zoonosis / transmission:** not applicable (genetic disease).

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15. Model Organisms

Mouse (primary model). - **Type:** mammalian, germline **knockout** — *Bgn*-deficient (null) mice (X-linked, so hemizygous males "Bgn0/Y"). Also **Bgn/Fmod double-knockout** for skeletal studies. - **Cardiovascular phenotype recapitulation (strong):** "*50% of biglycan-deficient male mice died suddenly within the first 3 months of life... aortic rupture that involved an intimal and medial tear as well as dissection between the media and adventitia*"; aortas showed "*structural abnormalities of collagen fibrils and reduced tensile strength*" (P 17502576). This closely models the human early-onset, male-predominant TAAD. - **Skeletal phenotype recapitulation:** *Bgn*-null mice develop "*an osteoporosis-like phenotype*" with reduced bone mass increasing with age (P 9731537), and reduced osteoblast BMP-4 responsiveness/differentiation (P 15173106) — models the skeletal component. - **Model limitations:** aortic-rupture penetrance is **strain/background-dependent** (prominent on BALB/cA; P 17502576); craniofacial/vascular-tree features less characterized; TGF-β activation dynamics in mouse aorta not fully mapped to human. - **Genetic model variants available:** single *Bgn* KO; *Bgn/Fmod* and *Bgn/Dcn* compound mutants (MGI). Conditional/humanized *BGN* lines are not standard.

Cellular / in vitro models. - **Patient-derived iPSC line BBANTWi009-A** from an MRLS male with a *BGN* deletion — normal karyotype, pluripotent, tri-lineage differentiation, original genotype retained (P 36599284). Enables SMC/vascular disease modeling and drug screening. - **Patient skin fibroblasts** used for cDNA/Western LoF confirmation (P 38531898).

Resources. MGI (mouse *Bgn*), IMSR (strains), Cellosaurus/biobank (iPSC line), ClinVar (human variants).

Supported vs. Refuted Hypotheses

Supported. - MRLS is X-linked, caused by *BGN* loss-of-function (P 27632686; 38531898). - Mechanism = ECM/collagen weakening + increased TGF-β/SMAD2 signaling (P 27632686; 17502576; 8038266). - Male-predominant severity/penetrance; disease affects the whole arterial tree (P 38531898).
- *Bgn*-null mouse recapitulates aortic dissection and skeletal phenotype (P 17502576; 9731537).

Refuted / not supported. - Pathogenic **missense-only** *BGN* variants driving disease via dominant-negative mechanism — **not supported**; LoF is the operative mechanism (P 38531898).
- A primary metabolic/enzymatic defect — not applicable; defect is structural-ECM/signaling.

Limitations and Future Directions

- Evidence rests on **small cohorts** (≈18 probands); precise phenotype frequencies, penetrance quantification, and natural-history/survival statistics are lacking.
- **No MRLS-specific clinical trials**; medical therapy (β-blocker/ARB) efficacy is extrapolated, not proven.
- Whether TGF-β hyperactivation is a **primary driver vs. secondary response** is unresolved.
- Female expressivity determinants (XCI skewing) and modifier genes (beyond *ATP2B3*) are under-characterized.
- **Future work:** iPSC-SMC and improved mouse models to define causal chain and test targeted (anti-TGF-β, ARB) and gene-based therapies; registry-based natural-history and QoL studies; validated circulating biomarkers.

Key ontology quick-reference

- **Disease:** MONDO:0010515; OMIM:300989; ORPHA:404003
- **Gene/protein:** HGNC:1044 (*BGN*); UniProt P21810; GO:0005518, GO:0050431, GO:0030198, GO:0007179
- **Phenotypes (HPO):** HP:0002647 (aortic dissection), HP:0002616 (aortic root aneurysm), HP:0004942 (arterial aneurysm), HP:0005116 (arterial tortuosity), HP:0000316 (hypertelorism), HP:0000766 (pectus deformity), HP:0001382 (joint hypermobility), HP:0001371 (contractures), HP:0002650 (scoliosis)
- **Anatomy (UBERON):** UBERON:0000947 (aorta), UBERON:0035904 (ascending aorta), UBERON:0001637 (artery), UBERON:0000948 (heart)
- **Cells (CL):** CL:0000359 (vascular smooth muscle cell), CL:0000057 (fibroblast), CL:0000062 (osteoblast)
- **Chemicals (CHEBI):** CHEBI:6541 (losartan), CHEBI:2719 (angiotensin II)
- **Treatments (NCIT):** C61912 (losartan), C2019 (β-blocker), aortic aneurysm repair (surgical)
